

PARSHAN PAKIMAN

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🔄 May, 2026

APPOINTMENTS

Assistant Professor of Operations Management

Since Fall 2025

Department of Operations Management and Strategy
University at Buffalo School of Management
The State University of New York (SUNY)

Principle Researcher

2023-2025

Tolan Center for Healthcare
Booth School of Business
The University of Chicago

EDUCATION

College of Business Administration, University of Illinois Chicago

2017 - 2023

Ph.D. Information and Decision Sciences

Thesis title Self-guided Approximate Linear Programs: Randomized Multi-shot Approximation of Markov Decision Processes ([Link](#))

Committee Selva Nadarajah (Chair); Dan Adelman, Beryl Chen, Itai Gurvich, Negar Soheili

College of Business Administration, University of Illinois Chicago

2017 - 2023

M.Sc. Business Analytics

College of Science, University of Tehran

2012 - 2016

B.Sc. Applied Mathematics

RESEARCH INTERESTS

- **Self-Adapting Approximate Dynamic Programming:** My research advances frameworks for dynamic decision making under uncertainty that are accessible to non-expert users. Rather than requiring users to manually design models, tune parameters, and solve large-scale optimization problems through trial and error, these frameworks automatically design, compute, and enhance their own approximations and produce high-quality decisions. By self-adapting to each problem instance's data and using information revealed during computation, these methods substantially reduce deployment complexity while maintaining strong theoretical performance guarantees, making advanced decision-support tools more accessible to users with limited domain expertise.
- **AI-Optimization Hybrids for Modern Business Applications:** My research investigates emerging operations management problems shaped by real-world operational complexities, including operating-room scheduling systems that account for evolving workforce composition, pricing and demand-learning mechanisms that maintain consistency across retail channels, and marketing campaigns that account for long-term and cross-store effects. To address these challenges, I develop tailored algorithms that integrate AI and data-driven optimization technologies to deliver high-quality decision support and generate managerial insights for these complex dynamic systems.

PUBLISHED PAPERS

- P. Pakiman, S. Nadarajah, N. Soheili, Q. Lin. *Self-Guided Approximate Linear Programs: Randomized Multi-Shot Approximation of Discounted Cost Markov Decision Processes*. **Management Science**, 2025 ([Link](#)).
- A. Chenreddy, P. Pakiman, S. Nadarajah, R. Chandrasekaran, R. Abens. *SMOILE: A Shopper Marketing Optimization and Inverse Learning Engine*. **Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining**, 2019 (Acceptance rate 6.4%, [Link](#)).

WORKING PAPERS

- D. Adelman, A. J. Mersereau, P. Pakiman. *Dynamic Optimization of Workforce Talent*. Major revision; revised manuscript in preparation for second-round submission to **Operations Research** ([Link](#)).
- P. Pakiman, B. Chen, S. Nadarajah, S. Jasin. *Adaptive Risk Mitigation in Demand Learning*. Under the first round of review at **Manufacturing & Service Operations Management** ([Link](#)).

- A. Cire, S. Nadarajah, P. Pakiman, N. Soheili. *Self-Adapting Approximations of Markov Decision Processes: A Guided Tour*. Under the first round of review at **INFORMS TutORials in Operations Research** ([Link](#)).
- P. Pakiman, S. Nadarajah. *Back to the Future: Revisiting a Pioneering Approximation of Average Cost Markov Decision Processes Using a Multi-Shot Perspective*. In preparation for submission to **INFORMS Journal on Computing** (Earlier version available at [Link](#)).

INDUSTRY COLLABORATIONS

- Interned with the Advanced Solutions team at *Guidehouse*: Developed an approximate dynamic programming algorithm based on simulation to address a workflow scheduling problem. 2021
- Collaborated with *Foresight ROI*: Developed an inverse optimization method using Markov decision processes to model historical data trajectories and optimize future marketing campaigns. ([Link](#) to the research paper). 2017 - 2019

TEACHING EXPERIENCES

Instructor

Since 2025

- Supply Chain Analytics (UB; [Link](#))
- Production & Operations Management (UB; [Link](#))

Guest Instructor / Teaching Assistant

2014-2023

- Approximate Dynamic Programming (UChicago)
- Supply Chain Management (UIC).
- Introduction to Operations Management (UIC)
- Business Analytics (UIC)
- Business Data Mining (UIC)
- Online Customer Analytics (UIC)
- Text Analytics for Business (UIC)
- Business Forecasting (UIC)
- Optimization for Analytics (UIC)
- Scientific Computing (UTehran)
- Numerical Linear Algebra (UTehran)

INVITED TALKS AND CONFERENCE PRESENTATIONS

Dynamic Optimization of Workforce Talent

- INFORMS Advances in Decision Analysis (ADA) Conference, Durham, NC, USA 2026
- INFORMS MSOM Conference, Cambridge, MA, USA 2026
- IMSI Workshop on Advances in Quantitative Medical Care, Chicago, IL, USA 2026
- INFORMS Applied Probability Society (APS) Conference, Atlanta, GA, USA 2025
- University at Buffalo, School of Management, Buffalo, NY, USA 2024
- Stony Brook University, College of Business, Stony Brook, NY, USA 2024
- University of Toronto, Rotman School of Management, Toronto, ON, Canada 2024
- Purdue Operations Conference, West Lafayette, IN, USA 2024
- INFORMS MSOM Conference, Minneapolis, MN, USA 2024

Adaptive Risk Mitigation in Demand Learning

- Canadian Operational Research Society (CORS) Annual Conference, Kingston, ON, Canada 2026
- POMS Annual Conference, Reno, NV, USA 2026
- POMS Annual Conference, Atlanta, GA, USA 2025
- INFORMS Annual Meeting, Phoenix, AZ, USA 2023
- INFORMS Annual Meeting, Virtual 2020
- INFORMS Revenue Management and Pricing (RMP) Student Live Paper Series, Virtual 2020

Self-Guided Approximate Linear Programs: Randomized Multi-Shot Approximation of Discounted Cost Markov Decision Processes

- Tuck School of Business, Dartmouth College, Hanover, NH, USA 2022
- International Conference on Continuous Optimization (ICCOPT), Bethlehem, PA, USA 2022

- INFORMS Optimization Society (IOS) Conference, Greenville, SC, USA 2022
- INFORMS Annual Meeting, Anaheim, CA, USA 2021
- POMS Annual Conference, Washington D.C., USA 2019
- INFORMS Annual Meeting, Phoenix, AZ, USA 2018
- POMS Annual Conference, Houston, TX, USA 2018

SMOILE: A Shopper Marketing Optimization and Inverse Learning Engine

- International Conference on Knowledge Discovery & Data Mining (ACM SIGKDD), Anchorage, AK, USA 2019

SERVICE

Ad-hoc Reviewer

- Operations Research (OR)
- Production and Operations Management (POM)
- INFORMS Journal on Computing (IJOC)
- Information Systems Research (ISR)
- International Conference on Learning Representations (ICLR)
- Annals of Operations Research (ANOR)
- Computers & Operations Research (COR)
- Electronic Commerce Research (ECR)
- Information Systems and Operational Research (INFOR)
- Manufacturing and Service Operations Management (MSOM) Conference

Conference Organization

- Session co-chair, *Data-driven Decision Making in Supply Chain*, POMS Annual Conference 2026
- Session chair, *Dynamic Pricing and Risk Management*, POMS Annual Conference 2025
- Session co-chair, *New Topics in Dynamic Assignment*, INFORMS Annual Meeting 2024
- Session co-chair, *Recent Advances in Approximate Dynamic Programming*, INFORMS Annual Meeting 2024
- Session co-chair, *Risk Management in Reinforcement Learning*, INFORMS Annual Meeting 2023
- Session co-chair, *Learning and Sequential Decision Making*, INFORMS Annual Meeting 2022
- Session co-chair, *Large-scale Linear Programs and Applications*, INFORMS Optimization Society Conference 2022
- Session chair, *Recent Advances in Reinforcement Learning*, INFORMS Annual Meeting 2021
- Session co-chair, *Social Responsibility and Risk in Supply Chains*, INFORMS Annual Meeting 2021

Academic Membership

- Production and Operations Management Society (POMS) 2018 - Present
- Institute for Operations Research and the Management Sciences (INFORMS) 2017 - Present
- INFORMS Chicago Chapter Ambassador 2022

COMPUTATIONAL SKILLS

- GitHub repositories: Self-Adapting MDP Approximations: [Link](#)
- Multi-Shot Approximation of MDPs: [Link](#)
- Programming languages: Python, R, C++, C, Java, MATLAB, HTML, JavaScript
- Machine learning: Scikit-learn, Gymnasium, Numba, Matplotlib, Pandas, PyTorch
- Optimization: Gurobi, CVXPY, Pyomo, OR-Tools, Nevergrad, Autograd
- Operating systems: Linux, macOS